servers directly, they get duplicated authorization logic, inconsistent state, and fragile orchestration.

An MCP gateway is a single, AI-aware proxy. The LLM app connects to the gateway once. The gateway knows how to talk to every registered MCP server, and it provides a stable, consistent interface back to the agent. Figure 12.9 is a simple visualization of the MCP gateway concept.

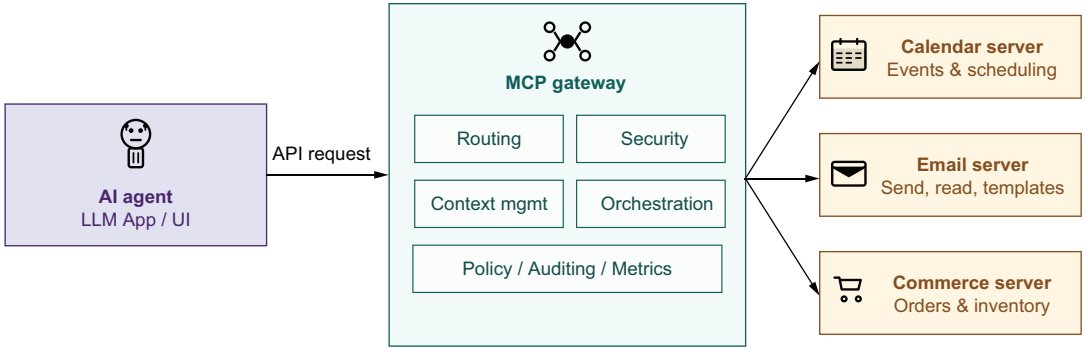


Figure 12.9 An MCP gateway centralizes how an AI agent connects to multiple MCP servers, handling routing, security, and context so that the application sees a single unified interface.

The MCP gateway reduces the complexity that arises when an LLM-powered web app must connect directly to many independent MCP servers. Instead of duplicating authentication and juggling inconsistent states, the application connects once to the gateway, which handles security, routing, and context on its behalf. This allows the app to treat all services as part of a single unified interface. Because the gateway maintains shared context, multistep workflows feel seamless; for example, scheduling a meeting can also trigger an email confirmation and a CRM update without the agent managing each integration separately. In this way, the gateway functions less like a traditional API proxy and more like an AI-aware service mesh, centralizing trust and enabling smoother orchestration across tools.

12.6.2 MCP-as-a-service

MCP-as-a-service is emerging as a natural evolution for businesses looking to integrate with AI assistants. Just as SaaS transformed software delivery by centralizing functionality in the cloud, MCP endpoints allow companies to expose their services directly to LLM-powered applications. Instead of building custom integrations for each AI platform, a business can operate a single MCP server and let users connect it to any compatible assistant.

For example, a local bookstore could run an MCP server at `mybookstore.com/mcp`. Customers might click a button labeled Add This Store to OpenAI, granting the AI